

Group Project — Guidelines

Introduction to Machine Learning

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Description

This document defines the structure, timeline, and assessment principles of the group project for the course 'Introduction to Machine Learning'. The project follows a progressive, Problem-Based Learning (PBL) approach and is designed to support learning through iteration and feedback.

Project Philosophy

The group project is conceived as a learning process rather than a single final deliverable. Students are expected to make informed decisions, reflect on limitations, and adjust their approach based on feedback received during the semester.

To support this process, the project is divided into three deliverables. The first two are evaluated qualitatively, with formative feedback only. Only the final deliverable receives a quantitative grade.

Project Scope

Students are strongly encouraged to propose a machine learning problem of their choice that better reflects their interests. Please discuss your idea with the professor. The group is expected to apply suitable machine learning algorithms, either learned in class or self-taught. The problem under study should involve data science algorithms (classification, regression, clustering). The results should be presented in graphical/table formats, accompanied by the corresponding analysis, discussion and conclusions.

The use of AI tools is allowed; they should be mentioned in the final document, with preference given to those that enable reference tracking (e.g., scite.ai, consensus.app, elicit.com).

Groups Composition

Each group must be composed of three students enrolled in the same practical class.

Deliverable 1 — Problem Formulation

Submission date: 27/03/2026

Assessment type: Qualitative (formative feedback only)

Objective: To evaluate the group's ability to formulate a meaningful machine learning problem and to reason about data and evaluation choices.

Expected content:

- Group definition
- Context and motivation of the problem
- Dataset description (origin, limitations, potential bias)
- Type of ML problem (classification or regression)

- Initial risks and assumptions

Deliverable 2 – Preliminary Results

Submission date: 24/04/2026

Assessment type: Qualitative (formative feedback only)

Objective: To assess the maturity of the modeling strategy and the ability to critically reflect on technical decisions.

Expected content:

- Data processing methodology
- Models explored and justification for their selection
- Preliminary ethical considerations
- Adjustments made since Deliverable 1

Deliverable 3 – Final Project

Submission date: 29/05/2026

Assessment type: Quantitative

Expected content:

- Data processing pipeline
- Evaluation strategy (e.g., cross-validation, metrics)
- Discussion about overfitting and underfitting
- Responsible and ethical considerations

Components:

- Oral presentation
- Developed code
- Final report (mandatory: Critical Analysis - evidencing the students' reflection on the acquired knowledge; Acknowledgement of used AI tools – which tools had been used, the rationale behind the selection)

Assessment criteria:

- Clarity of problem formulation
- Coherence and justification of modeling decisions
- Appropriate evaluation and interpretation of results
- Reflection on limitations and ethical implications
- Quality of communication and structure

Project Timeline and Integration with Classes

Week 1–6: Project ideas are introduced informally. Students form groups and explore possible topics.

Week 7 (27/03/2026): Submission of Deliverable 1 (Problem Formulation).

Week 8–9: Technical consolidation. Groups refine their approach based on feedback.

Class 10 (24/04/2026): Submission of Deliverable 2 (Data exploration, Feature engineering, Modeling Strategy).

Classes 11–13: Supervised project development, ethical reflection, and interpretability discussion.

Class 14 (29/05/2026): Final presentation and submission of Deliverable 3.

Evaluation:

- Submission 1 and 2: No quantitative assessment, only qualitative feedback if adjustments are needed.
- Final Document: Evaluated based on document quality, technical/scientific accuracy, problem complexity, and critical thinking. Maximum of 6 pages, IEEE format.
- Presentation: Evaluated based on presentation quality and ability to discuss the topic.
 - Each group will have 10 min for presentation followed by 5 min for discussion
- Peer evaluation within the group.
- Evaluation by classmates.
- Document (40%), presentation (40%), Peer evaluation within group (10%), Classmate Evaluation (10%)

Goal:

Develop oral and written communication skills, critical thinking, logical reasoning, and discussion abilities.

Enhance proficiency in using machine learning tools, building processing pipelines, evaluating performance, and conducting a technical discussion of the results.

Important Notes

- Adjustments between deliverables are expected and encouraged.
- Early mistakes are not penalized if addressed later.
- The final grade reflects the overall maturity of the project, not only performance metrics.
- Ethical reflection and responsible use of ML are integral parts of the evaluation.

Suggested Data Repositories:

- Kaggle Data Repository : <https://www.kaggle.com/datasets>
- UCI Machine Learning Repository : <https://archive.ics.uci.edu/ml/index.php>

Suggested Document Structure:

The document should clearly reflect the work carried out, allowing for an understanding of how the data was processed and the entire methodology implemented. The results should be presented critically and based on evidence.

- **State of the art review**

Search and review of at least 5-6 references (papers, reports, thesis, etc.) handling the same or similar problem. Make a review of different techniques used to solve the problem you want to explore.

- **Data description, visualization and statistical analysis**

Describe the problem you want to solve, the features and visualize the data (if it is difficult due to high dimension, show only some samples). Provide some statistical analysis such as metadata (e.g. features range of variation), histograms, try to identify if there are some data quality problems, detect interesting subsets.

- **Data preprocessing (if relevant)**

Describe possible preprocessing steps to construct the final input to the machine learning algorithm from the initial data, such as data normalization, feature selection or dimensionality reduction in case of redundant features.

- **Description of the applied machine learning algorithm(s)**

Apply a suitable ML algorithm (learned in class or self-learned) to solve the problem with the chosen dataset. Introduce the method shortly, define its parameters. Make a selection of the most important model hyper parameters after their variation in a selected range. Show graphically the results of this search.

Clear statement what is the ML problem (classification, regression, what are inputs (how many), and the outputs of the model).

- **Presentation and discussion of results**

Presentation of the results preferably in a graphical format. Analysis, discussion, interpretation. Compare your results with the results in the reviewed references or apply and compare at least two ML methods on the same problem.

- **Conclusions**

Critical discussion of the gained knowledge regarding the advantages/disadvantages of the applied methods on the problem in hand. Describe the problem complexity. Suggestions for potential future directions of study.

- **Critical Reflection**

Critical description of the selected topic, evidencing the students' reflection on the acquired knowledge.

- **AI Tools**

Enumerate the AI tools used and the rationale behind the selection.